

(Đề thi thực hành trên EOS)

I Case Study

MedCare Solutions (MCS) is a mid-sized healthcare technology firm based in Vietnam whose mission is to enhance patient care through digital innovation. The company serves public and private medical clinics across the country. Seeing the rising demand for efficient healthcare management tools, MCS's board of directors (BOD) has decided to build a comprehensive **Clinic Management System (CMS)** to modernize clinic operations and completely replace paper-based workflows.

Through a single centralized platform, the CMS connects patients, doctors, nurses, receptionists, pharmacists, and administrators. When a patient arrives, a receptionist — or the patient themselves — creates a digital profile storing demographics, health insurance, allergies, and full medical history. Appointments can be booked online or at the front desk with real-time doctor availability, automated SMS/email reminders, and dynamic queue management with live wait-time estimates on the day of the visit. During the consultation, nurses record vital signs (BP, temperature, HR, SpO2) while doctors enter ICD-10 diagnoses and treatment plans via structured templates; all entries are time-stamped and attributed to the responsible clinician. Lab and imaging orders (X-ray, ultrasound, MRI) are raised electronically from the EMR, results are auto-linked to the patient record, and critical values trigger immediate alerts. Doctors issue e-prescriptions with automatic drug-drug interaction and allergy checks; prescriptions flow to the pharmacy for dispensing with real-time inventory updates and low-stock alerts. Administrators manage user roles (RBAC), schedules, and fee structures, and can view KPI dashboards covering patient volume, revenue, and staff productivity.

In addition to the above functional requirements, the system must allow staff users to sign in with MCS's corporate email account via the MCS Identity Platform (Single Sign-On). The CMS must be intuitive and easy to use, even for clinical staff with limited technical experience. The interface must be clean and consistent, with clearly labeled navigation for high-frequency tasks such as patient lookup, appointment creation, and prescription entry. Color coding must be used meaningfully — green to confirm actions, red for errors or critical alerts — so staff can act quickly without ambiguity.

The system must cope with peak transaction volumes without performance degradation, especially during morning rush hours when patient check-ins, consultations, and lab requests happen simultaneously. Under normal operating load, the system shall respond to user interactions (page loading, form submission, search) within 2 seconds. The system shall also scale horizontally to support a growing number of patients, clinics, and staff users without service interruption.

The CMS is a green-field project with evolving requirements; MCS's IT department leads the development with **3–4 experienced developers** and clinical representatives, targeting the first release in **3 months** and full completion within **9 months**.

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II Questions

Assume that you are the one assigned to implement this project. Read the description carefully and answer the following questions:

- 1. Your development team is made up of highly skilled, experienced members. Your manager proposes using the **XP framework** for this project. Do you agree or disagree? List and explain all main principles, practices, and characteristics of the **XP framework**, then match each one to the project characteristics above to justify your answer. (2.0 points)
- 2. Your manager requires the system to be tested from the smallest unit level up to the fully integrated system level. At the unit level, testing must focus on program structure; at higher levels, testing must be based on functional requirements.
 - a. What type and level/stage of testing does your manager require? (0.7 point)
 - b. Who performs testing at each stage/level, and why are they the most suitable for that role? (0.3 point)
- 3. Based on the project description above, identify and list: **5 functional requirements**, **3 performance requirements**, and **2 usability requirements**. (2.0 points)
- 4. Write **5 user stories** based on your answers in Question 3. (1.5 points)
- 5. The manager requires you to apply design patterns to the software system. List and describe in detail two design patterns discussed in the courses (1.0 points).
- 6. Build a **story map** for the "Manage Patient Appointments" activities described in the project above. (2.5 points)

III Notes

a) Mỗi câu trả lời không đúng sẽ bị trừ 0.1 điểm b) Dưới đây là hướng dẫn trình bày Story Map trên phần mềm EOS.

Khi trình bày Story Map, cấu trúc được tổ chức thành 2 phần. Phần 1 (A) mô tả các activities và user tasks. Phần 2 (B) mô tả các user story cần thực hiện cho mỗi lần bàn giao. Các phần được trình bày như sau:

A. Activities and User tasks

- 1. <Activity 1 name>
 - 1.1 <User task 1 of activity 1>
 - 1.2 <User task 2 of activity 1>
- 2. <Activity 2 name>
 - 2.1 <User task 1 of activity 2>
 - 2.2 <User task 2 of activity 2>

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B. Releases

Release 1-----

<Liệt kê các user story cho từng user task sẽ thực hiện trong lần bàn giao đầu tiên. Danh chỉ mục cho các user story tương ứng với các user task ở phần trên>

- 1.1.1 User story 1 of user task 1 in activity 1
- 1.1.2 User story 2 of user task 1 in activity 1
- 2.1.1 User story 1 of user task 1 in activity 2
- 2.1.2 User story 2 of user task 1 in activity 2

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Release #-----

<Liệt kê các user story cho từng user task sẽ thực hiện trong lần bàn giao #>

- 1.1.3 User story 3 of user task 1 in activity 1
- 2.1.3 User story 3 of user task 2 in activity 2

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Ví dụ: Khi mô tả Story Map có cấu trúc biểu đồ sau trên EOS



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[Hình ảnh: Sơ đồ Story Map mẫu (Steve Rogalsky, winnipegagilist.blogspot.com) gồm 4 activities (Organize Email, Manage Email, Manage Calendar, Manage Contacts) ở hàng trên cùng, mỗi activity có nhiều user tasks bên dưới (Search Email, File Emails, Compose Email, Read Email, Delete Email, Search by Keyword, Move Emails, Create and send basic email, Open basic email, Delete email, Create sub folders, Send RTF email, Open RTF email, View Calendar, Create Appt, Update Appt, View Appt, Create Contact, Update Contact, Delete Contact, ...). Các tasks được nhóm theo 3 Releases: Release 1, Release 2, Release 3.]

Story Map này được mô tả trong EOS như dưới đây.

A. Activities and User Tasks

1. Organize email
 - 1.1 Search email
 - 1.2 File email
2. Manage email
 - 2.1 Compose email
 - 2.2 Read email
 - 2.3 Delete email
3. Manage Calendar
 - 3.1 View Calendar
 - 3.2 Create Appt
 - 3.3 Update Appt
 - 3.4 View Appt
4. Manage contacts

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B. Releases

Release 1-----

- 1.1.1 Search email by key word
- 1.2.1 Move emails
- 1.2.2 Create subfolders
- 2.1.1 Create and send basic email
- 2.1.2 Send RTF email
- 2.2.1 Open basic email
- 2.2.2 Open RTF email

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Release 2 -----

- 1.1.2 Limit search to one field
- 1.1.3 Limit search to +one field
- 2.1.3 Send HTML email
- 2.1.4 Set email priority

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